About this PDF

Built only from: **NOC:Natural Language Processing (NPTEL course book, weeks 3, 4)** — https://nptel.ac.in/courses/106105158

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Covers 6 of 9 syllabus sub-topics of this unit.

Partly covered in the source:

- Stochastic and Transformation-based tagging
- Issues in PoS tagging

Not in the source (not covered in this PDF):

- Unsmoothed N-grams
- Evaluating N-grams, Smoothing
- Interpolation and Backoff

Word Classes

Definition

Word classes, also called part-of-speech categories, group words into grammatical types such as nouns, verbs, adjectives, adverbs, pronouns, determiners, and prepositions. They are divided into **open class** (content words) and **closed class** (function words) categories.

Explanation

Open-class words are called “open” because new words can continually be added to the language (e.g., Google, Photoshop, Skype) and they become nouns or verbs as they enter usage. Closed-class words form a fixed set (pronouns, determiners, prepositions, connectives) and serve to tie concepts together in a sentence.

Example

- Nouns: *chair*, *bandwidth*, *pacing*
- Verbs: *study*, *debate*, *munch*
- Adjectives: *purple*, *tall*, *ridiculous*
- Adverbs: *unfortunately*, *slowly*
- Pronouns: *I*, *me*, *mine*
- Determiners: *the*, *a*, *that*, *those*

Key points

- Open-class words are content words that can expand with new lexical items.
- Closed-class words are a fixed set of functional words that connect sentence elements.
- Typical open-class categories include nouns, verbs, adjectives, and adverbs.
- Typical closed-class categories include pronouns, determiners, and prepositions.

Part-of-Speech Tagging

Definition

Part-of-speech tagging is the problem of, given a set of words (a sentence or document), identifying the actual category for each word and providing a unique answer for each one.

Explanation

Tagging schemes can be coarse (e.g., only noun, verb, adjective, adverb) or finer-grained, providing grammatical details such as singular vs. plural noun, verb tense, or proper noun. A word may have multiple possible tags, and the correct tag often depends on the surrounding context. To resolve this ambiguity, one can use rule-based methods that look for contextual cues, or statistical methods that learn from a training corpus where each word is already tagged.

Example

The word **back** illustrates ambiguity: it is an adjective in “the back door”, a noun in “on my back”, an adjective in “win back”, and a verb in “promised to back the bill”.

Algorithm / Steps

1. Identify all words that can have multiple tags.
2. For each such word, find a contextual feature that helps decide between the possible tags.
3. If a new context contains that feature, assign the corresponding tag; otherwise, use a statistical model trained on a tagged corpus.
4. In statistical tagging, train on a corpus where each word is paired with its true tag, then learn a model that predicts the most probable tag given the context.

Key points

- Tagsets range from very coarse to fine-grained, and the choice depends on the desired grammatical detail.
- Ambiguity is common: about 40 % of word tokens are ambiguous in a large corpus.
- Rule-based approaches rely on contextual cues, while statistical approaches learn probabilities from annotated data.
- The tagging problem is essentially a disambiguation task that requires assigning a single, correct tag to each word in context.

Rule-based

Definition

A rule-based approach solves NLP tasks by writing explicit “if-then-else” patterns that capture linguistic regularities, such as deciding which part-of-speech (POS) tag to assign to a word based on its context.

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Explanation

The method relies on knowledge about a word's typical POS distribution and the local context in which it appears. For example, determiners rarely follow each other, and a determiner is usually followed by an adjective or a noun; such constraints can be encoded as rules. By examining a training corpus, one can extract or manually craft transformation rules that modify an initially guessed tag when certain contextual conditions are met.

Example

Rule: **MD → NN** when the preceding word is a determiner (DT). In a new sentence, if a word is initially tagged as MD and the previous token is DT, the rule changes the tag to NN.

Algorithm / Steps

1. For each word in the training corpus, assign the most probable POS tag based on overall frequency.
2. Compare these initial tags with the gold-standard tags in the corpus.
3. For every mismatch, formulate a transformation rule that corrects the tag under a specific context (e.g., "if previous word is DT, change MD to NN").
4. Evaluate the rule set on a separate test set to measure tagging accuracy.

Key points

- Rule-based POS tagging uses explicit conditional patterns derived from linguistic insight or corpus analysis.
- Contextual information (previous/following words) is essential for disambiguating ambiguous words.
- Transformation-based learning (TBL) creates rules that modify an initial most-likely tag to match the correct tag observed in training.
- Example rule: change MD to NN when preceded by a determiner (DT).

Stochastic and Transformation-based tagging

Partly covered in the source.

Definition

Transformation-based learning (TBL) is a method that learns transformation rules from a tagged training corpus to correct the most probable part-of-speech tags assigned to words in new sentences.

Example

- change **MD** to **NN** (when preceded by a determiner)
- change **VBD** to **VBN**

Algorithm / Steps

1. Start with a training corpus where each word is annotated with its true part-of-speech tag.

2. For every word, assign the most probable tag based on simple frequency statistics.
3. Compare the assigned tags with the true tags in the training data.
4. For each mismatch, generate a transformation rule that changes the incorrect tag to the correct one, using contextual cues (e.g., preceding determiner).

Key points

- Initial tags are chosen as the most likely tags for each word.
- Rules are based on contextual information such as neighboring words or tags.

Issues in PoS tagging

Partly covered in the source.

Definition

Issues in part-of-speech (PoS) tagging arise when the modeling assumptions are too restrictive to capture the necessary contextual information for accurate tag assignment.

Explanation

Standard hidden Markov models (HMMs) adopt a first-order Markov assumption, meaning each tag is conditioned only on the immediately preceding tag. This limited context can fail to disambiguate words that require broader syntactic or lexical cues. For instance, the word “marked” can be a past participle in one sentence and a past-tense verb in another, yet the previous tag alone does not provide enough information to choose correctly. To address this, one may employ higher-order HMMs (e.g., second-order) that consider a longer tag history, or shift to discriminative models such as maximum entropy, which allow incorporation of heterogeneous features beyond just the previous tag or word.

Example

The sentence pair illustrates the problem:

1. “… is clearly **marked** …” (past participle)
2. “… he **marked** …” (past-tense verb) – the preceding tag does not suffice to differentiate the two uses.

Key points

- First-order HMMs rely only on the previous tag, limiting contextual awareness.
- Such models may mis-tag words that need broader syntactic cues (e.g., “marked”).
- Higher-order HMMs or discriminative models (maximum entropy) can incorporate additional features to overcome these limitations.
- Maximum entropy models are discriminative and permit a heterogeneous set of features, not just tag-word pairs.

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Hidden Markov and Maximum Entropy models.

Definition

Hidden Markov models are generative models used for sequence labeling such as part of speech tagging, while maximum entropy models are discriminative models that allow a heterogeneous set of features to be combined in a probabilistic framework for the same task.

Explanation

HMMs rely on the first order Markov assumption, where each tag depends only on the previous tag, which limits their ability to use richer contextual information such as word morphology or distant words. Maximum entropy models do not make additional assumptions beyond the observed constraints; they choose the most uniform distribution that satisfies the feature constraints derived from the data, thereby maximizing entropy. In a maximum entropy model the probability of a class y given an observation x is computed as an exponential of a weighted sum of feature functions f i (x, y) and a normalizing constant that ensures the probabilities sum to one.

Example

A feature could be “the word is the first word of the article”, “the next word is ‘to’”, or “one of the previous five words is a preposition”; such features can be included in a maximum entropy model but not in a standard HMM.

Key points

- HMMs use only the previous tag as context, which can miss important cues such as morphology or distant words.
- Maximum entropy models are discriminative and can incorporate any set of features that may influence the tag decision.
- The model selects the distribution with maximum entropy among those that satisfy the observed feature constraints, leading to the most uniform solution consistent with the data.
- Probability computation in a maximum entropy model follows the form: exponent of sum (lambda i * f i (x, y)) normalized over all possible y.

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Changes made by AcademicsGen: the source text was rephrased, reorganised and grammar-corrected into study material; nothing added.

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